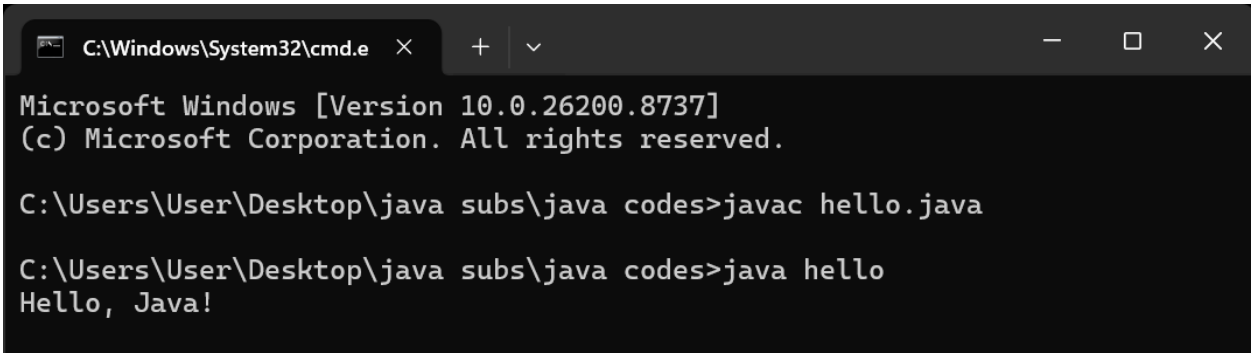
List of Java Programming Lab Programs based on Basic Java Concepts, Control Statements, Java Keywords, Primitive Data Types, Variables, Type Conversion, and Type Casting.

Program 1: Hello Java
Aim:
To write and execute a Java program that displays "Hello, Java!" on the console.
Procedure:

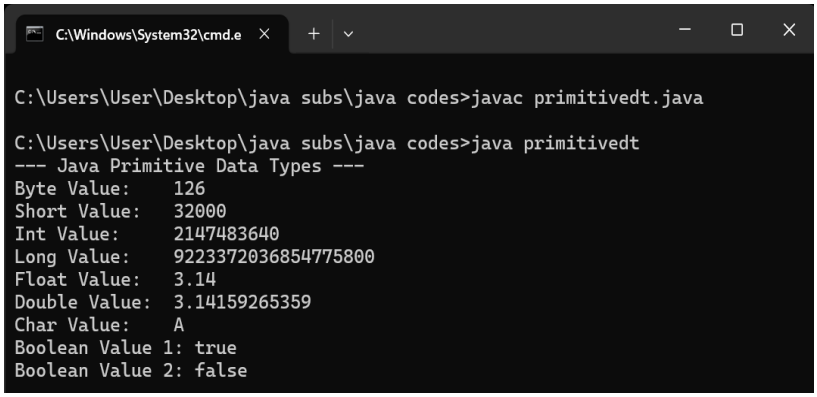
1. Create a Java file named Hello.java.
2. Write the program to print "Hello, Java!".
3. Compile the program using the javac command.
4. Execute it using the java command.
5. Observe the output.



Result:
The program was executed successfully and displayed "Hello, Java!" on the screen.

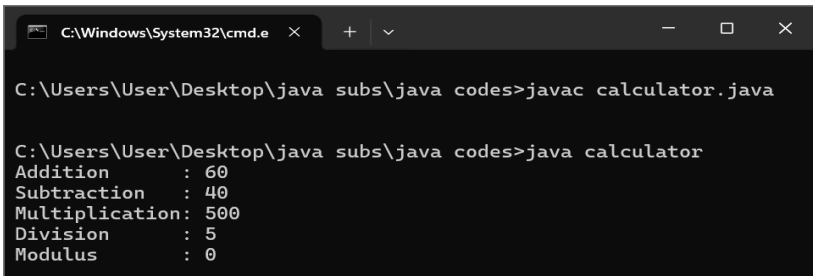
Program 2: Primitive Data Types
Aim:
To declare variables of different primitive data types and display their values.
Procedure:

1. Declare variables of type byte, short, int, long, float, double, char, and boolean.
2. Assign suitable values to each variable.
3. Display the values using output statements.
4. Compile and execute the program.



Result:
The values of all primitive data types were displayed successfully.

Program 3: Arithmetic Operations on Integers
Aim:
To perform arithmetic operations on two integer variables.
Procedure:

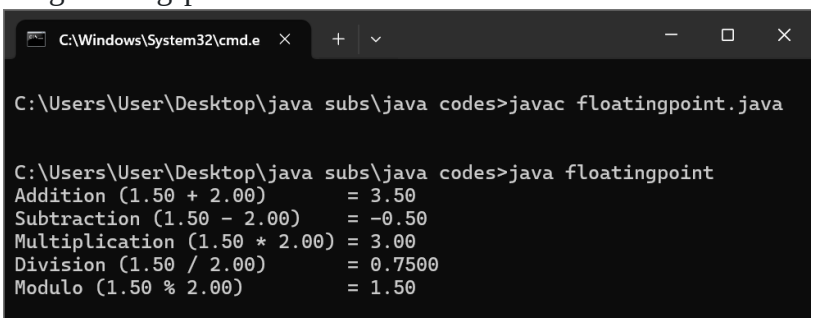


1. Declare two integer variables.
2. Perform addition, subtraction, multiplication, division, and modulus operations.
3. Display the results.
4. Compile and run the program.

Result:
The arithmetic operations on integer variables were performed successfully.

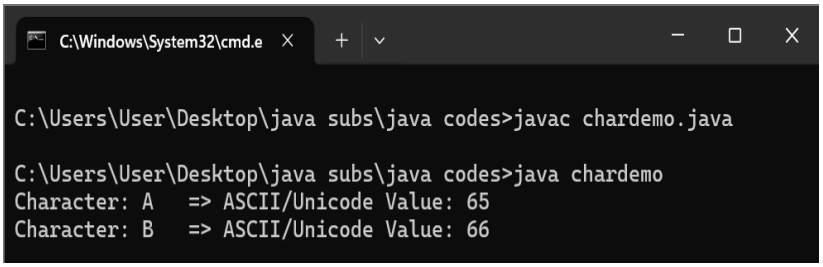
Program 4: Arithmetic Operations on Floating-Point Numbers
Aim:
To perform arithmetic operations using floating-point numbers.
Procedure:

1. Declare two floating-point variables.
2. Perform basic arithmetic operations.
3. Display the results.
4. Compile and execute the program.



Result:
The arithmetic operations on floating-point numbers were executed successfully.

Program 5: Character and ASCII/Unicode Value
Aim:
To display a character and its corresponding ASCII/Unicode value.
Procedure:

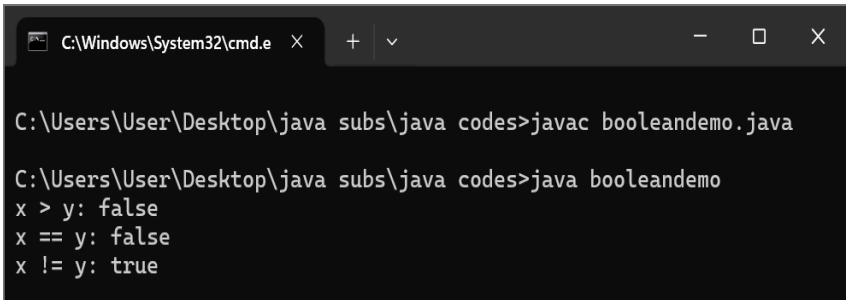


1. Declare a character variable.
2. Convert the character to its ASCII/Unicode value.
3. Display both the character and its value.
4. Compile and run the program.

Result:
The character and its corresponding ASCII/Unicode value were displayed successfully.

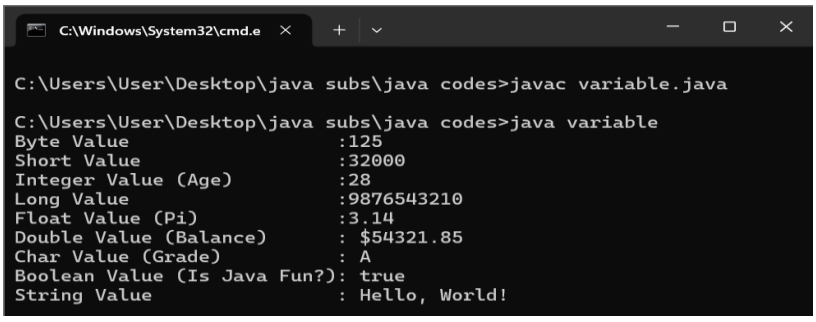
Program 6: Boolean Variables
Aim:
To demonstrate the use of Boolean variables and Boolean expressions.
Procedure:

1. Declare Boolean variables.
2. Perform simple Boolean expressions.
3. Display the results.
4. Compile and execute the program.



Result:
The Boolean values and expressions were displayed successfully.

Program 7: Variables
Aim:
To declare, initialize, and display different types of variables.

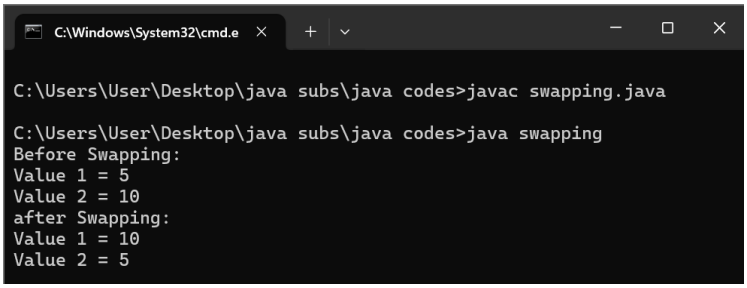


Procedure:
1. Declare variables of different data types.
2. Initialize them with suitable values.
3. Display the values.
4. Compile and execute the program.

Result:
The values of different variables were displayed successfully.

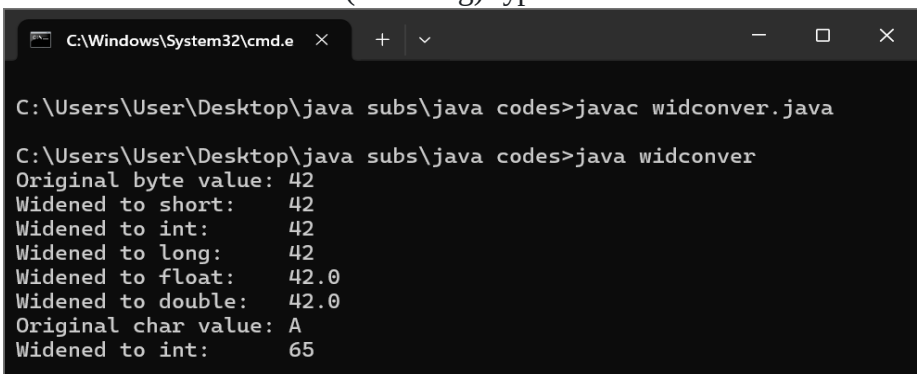
Program 8: Swapping Two Variables
Aim:
To swap the values of two variables using a temporary variable.

Procedure:
1. Declare two variables.
2. Use a temporary variable to swap their values.
3. Display the values before and after swapping.
4. Compile and run the program.



Result:
The values of the two variables were swapped successfully.

Program 9: Widening Type Conversion
Aim:
To demonstrate automatic (widening) type conversion.



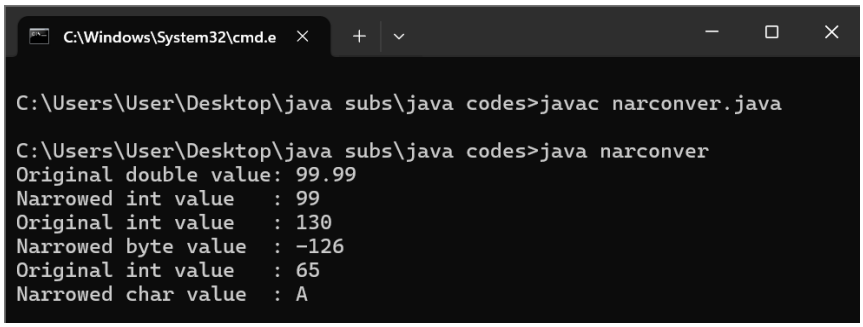
Procedure:
1. Declare variables of smaller data types.
2. Assign them to larger data types.
3. Display the converted values.
4. Compile and execute the program.

Result:
Automatic type conversion was performed successfully.

Program 10: Narrowing Type Casting
Aim:
To demonstrate explicit (narrowing) type casting.

Procedure:

1. Declare a variable of a larger data type.
2. Convert it to a smaller data type using type casting.
3. Display the converted value.
4. Compile and run the program.



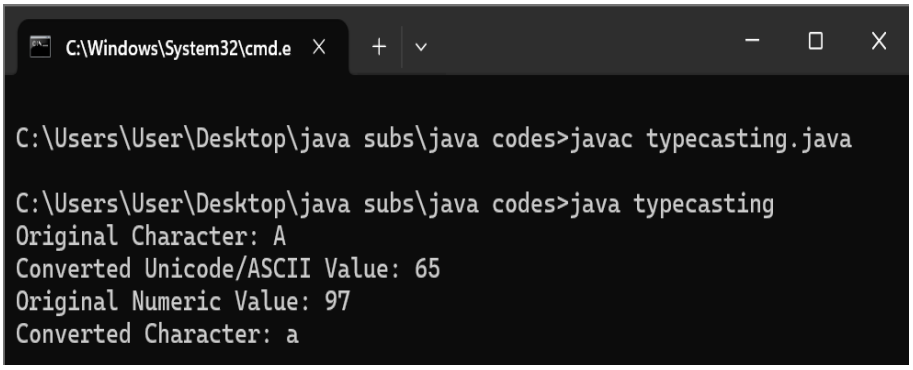
Result:

Explicit type casting was demonstrated successfully.

Program 11: Character to ASCII/Unicode Conversion

Aim:

To convert a character to its ASCII/Unicode value and vice versa.



Procedure:

1. Declare a character and an integer value.
2. Perform type casting between character and integer.
3. Display the converted values.
4. Compile and execute the program.

Result:

The character and ASCII/Unicode conversions were performed successfully.

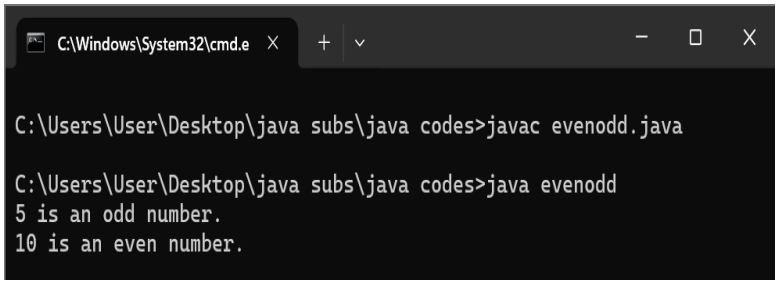
Program 12: Even or Odd

Aim:

To check whether a given number is even or odd using the if-else statement.

Procedure:

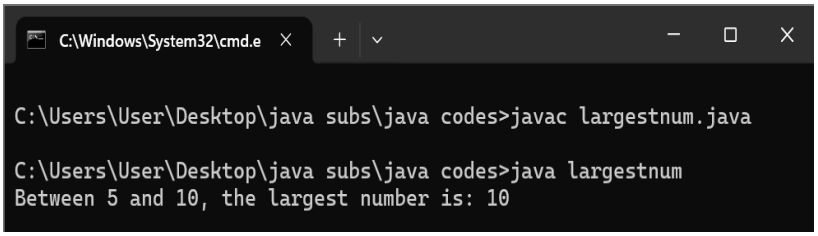
1. Declare an integer variable.
2. Use an if-else statement to check whether the number is even or odd.
3. Display the result.
4. Compile and run the program.



Result:

The program correctly identified whether the number was even or odd.

Program 13: Largest of Two Numbers
Aim:
To find the largest of two numbers using the if-else statement.

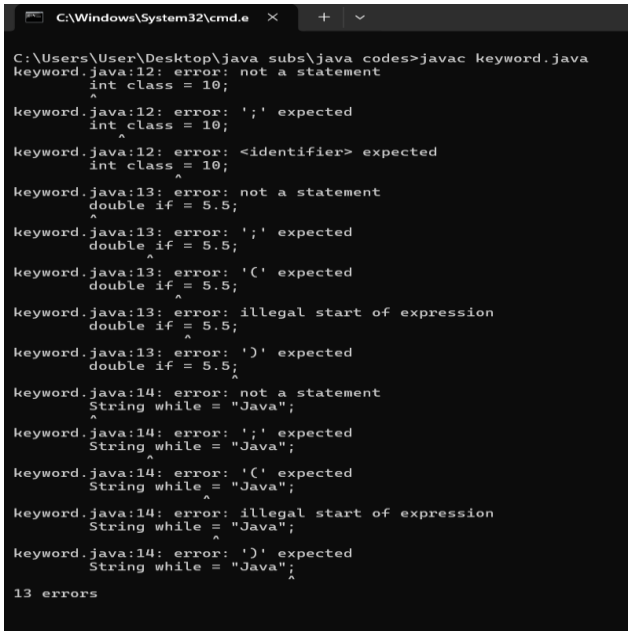


Procedure:
1. Declare two integer variables.
2. Compare them using if-else statements.
3. Display the larger number.
4. Compile and execute the program.

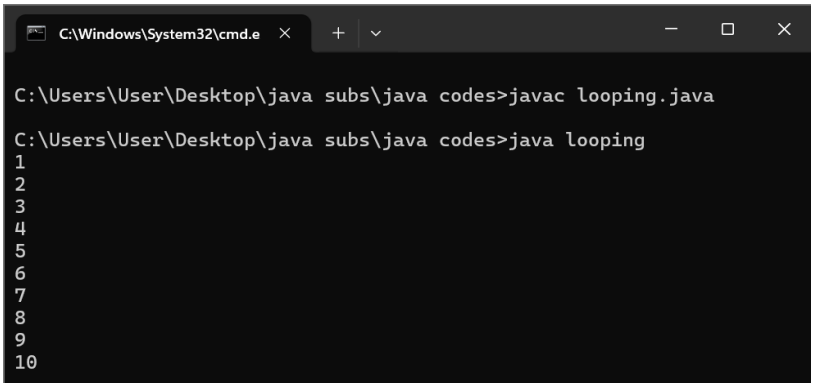
Result:
The larger of the two numbers was displayed successfully.

Program 14: Java Reserved Keywords
Aim:
To demonstrate the use of Java reserved keywords and understand why they cannot be used as variable names.
Procedure:
1. Create valid variable names.
2. Observe that Java keywords cannot be used as identifiers.
3. Compile and execute the program.
4. Note the output or compiler behavior.

Result:
The program demonstrated the use of valid identifiers and the restriction on Java keywords.



Program 15: For Loop
Aim:
To demonstrate the use of the for loop by displaying numbers from 1 to 10.



Procedure:

1. Write a for loop to iterate from 1 to 10.
2. Display each number.
3. Compile and execute the program.
4. Observe the output.

Result:

The numbers from 1 to 10 were displayed successfully.